

◆ Gemini

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA

# Read multiple images
images = []

# Assuming there is a single image file named 'image.jp
# and the intent is to process this image (or replicate
# The original code expected multiple numbered images w
# We will use the existing '/content/image.jpg' for loa

for i in range(1, 6):
    # Using the single available image file 'image.jpg'
    img_path = "/content/image.jpg"
    img = cv2.imread(img_path,
                     cv2.IMREAD_GRAYSCALE)

    if img is None:
        print(f"Warning: Could not load image '{img_pat
        # If the single image 'image.jpg' itself cannot
        # In a real scenario, you might want to handle
        break # Exit the loop if the primary image cann

    img = cv2.resize(img, (50, 50))
    images.append(img.flatten())

# Convert to NumPy array
# Ensure that the 'images' list is not empty before con
if not images:
    print("Error: No images were loaded. Cannot proceed
else:
    data = np.array(images)

    print("Original feature dimension:" data.shape[1])
```

```
# PCA with 95% information preservation
pca = PCA(n_components=0.95)

reduced_data = pca.fit_transform(data)

print("Reduced feature dimension:", reduced_data.shape[1])

print("Explained variance ratio:")
print(pca.explained_variance_ratio_)

print("Total information preserved:",
      sum(pca.explained_variance_ratio_))

# Reconstruct images
reconstructed = pca.inverse_transform(reduced_data)

# Display reconstructed images
plt.figure(figsize=(12, 4))

for i in range(len(images)):
    plt.subplot(1, 5, i + 1)

    img = reconstructed[i].reshape(50, 50)

    plt.imshow(img, cmap="gray")
    plt.title("Image " + str(i + 1))
    plt.axis("off")

plt.tight_layout()
plt.show()
```

Original feature dimension: 2500

Reduced feature dimension: 1

Explained variance ratio:

[nan]

Total information preserved: nan

/usr/local/lib/python3.12/dist-packages/sklearn/decomposition/_pca.py:648: RuntimeWarning: invalid value encountered in

$\text{explained_variance_ratio_} = \text{explained_variance_} / \text{total_var}$

Image 1

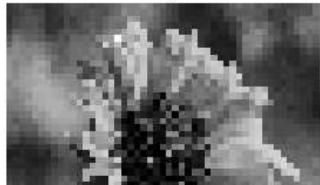


Image 2

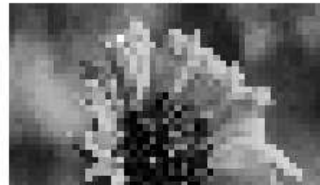


Image 3

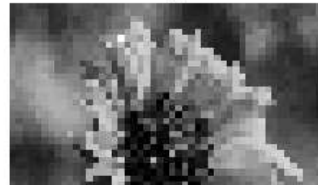


Image 4

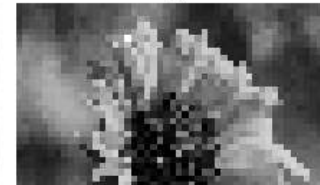


Image 5

